

Université Lumière Lyon 2

M2 Interactions Homme-Machines (IHM)

**Influence of automation level on cortical activation
across the take-over cycle: an fNIRS study**

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Abstract

Highly automated driving requires drivers to occasionally resume manual control upon a take over request (TOR), a transition that fundamentally reshapes the driver's cognitive state. While the neural substrates of motor control, task switching, monitoring, and cognitive control have been extensively mapped in isolation, how these four functions are dynamically integrated across a complete take-over cycle remains poorly understood. The present study used functional near-infrared spectroscopy (fNIRS) to examine how the level of automation (LoA) modulates the task-evoked cortical hemodynamic response across three periods of interest (PoI): the nominal driving period (NDP), the take-over preparation period (TOPP), and the manual driving period (MDP). In a driving simulator, 34 participants were each exposed to four automation solutions spanning from full manual control to highly automated driving (LoA1–LoA4), and cortical activation was represented by HbO general linear model coefficients across 4 regions of interest related to 4 cognitive functions. Linear mixed-effects models revealed a significant PoI effect and a $\text{LoA} \times \text{PoI}$ interaction, indicating that automation's influence on cortical activation was period-specific rather than a constant shift. Contrary to our hypothesis, cognitive control activity following manual takeover was lower than during continuous driving and declined linearly as the preceding level of automation increased, with the switching region showing an attenuated parallel trend. Motor region activity also decreased after takeover but was insensitive to automation level, and the monitoring region showed no reliable effect. These findings indicate that automation's modulation of post-takeover neural activity is region-specific, and that post takeover cognitive control re-engagement is a graded recovered process reflecting the lingering cost of automation-induced disengagement.

Key Words

fNIRS, Automation, take over request, cognitive control, motor control, switching, monitoring

Déclaration de contribution

La définition de la question scientifique, le choix de l'approche générale, le choix de la méthodologie spécifique ainsi que sa mise au point ont été déterminés par Jordan Navarro, encadrant du stage, et moi-même, Shenglan Chen, au cours de discussions; Arnaud Fournel et Emmanuelle Reynaud ont apporté leur aide sur le plan technique. La recherche bibliographique, le recrutement des sujets, le test des sujets, le codage et l'analyse des données, l'interprétation des résultats et la formulation des conclusions, ainsi que la rédaction du mémoire et la production des tables et figures, ont été réalisés par moi-même. La programmation de l'expérience s'est appuyée sur un système de simulation de conduite et des séquences expérimentales préexistantes (Deniel et al., 2024).

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I. Introduction

1.1. Literary review

One of the last decade's technological trends clearly points towards automated driving. However, due to persistent technical limitations, fully autonomous driving (SAE Level 5) which is capable of handling all scenarios without human intervention remains years away from widespread deployment. In the near to mid-term, the dominant form of advanced driving automation is highly automated driving (SAE Levels 3-4, HAD), in which the automated system manages the complete dynamic driving task in the conditions where hardware and software of driving system is designed to handle, yet drivers are still expected to be able to occasionally resume manual control upon request (Navarro, 2019). This expectation is a critical safety challenge : the take-over request (TOR). Beyond purely behavioral measures, Human-Machine Cooperation (HMC) offers a valuable theoretical framework for understanding this transition, by emphasizing that automation fundamentally reshapes the human cognitive state (Hoc et al., 2009; Navarro et al., 2011). Accordingly, resuming manual control from an automated state is not a trivial handoff but a demanding cognitive transition for the driver in charge (Navarro, 2019). This transition can be structured around three temporal periods of interest (PoI): the automated driving phase preceding the TOR, the control transition preparation phase, and the subsequent manual driving resumption (Deniel et al., 2024). The cognitive challenges inherent to each of these periods can be understood through the comprehension of cognitive and motor control.

In cognitive neuroscience, Cognitive Control (CC) is conceptualized as a process of behavioral regulation that optimizes goal-directed behavior and counters automaticity, achieved by establishing proper mappings between external inputs, internal states, and output actions (Friedman & Robbins, 2022). CC is not a unitary mechanism, but rather a

collection of component processes including task switching, monitoring, planning, and inhibitory control (Diamond, 2013; Friedman & Miyake, 2017; Jurado & Rosselli, 2007). Neurobiologically, CC relies on the active maintenance of patterns of activity in the prefrontal cortex (PFC) that represent goals (Miller & Cohen, 2001). The PFC provides top-down bias signals to downstream brain structures, guiding neural activity to resolve response conflicts and direct attentional resources toward task-relevant information. At the network level, cognitive control is primarily subserved by two intrinsic networks: the cingulo-opercular network (CO), which maintains task-set representation and sustains tonic alertness across a task, and the frontoparietal network (FP), which supports trial-by-trial adaptive updating in response to changing task demands (Cohen et al., 2014, Dosenbach et al., 2007, Sadaghiani et D'Esposito, 2015) . In the context of driving, the effective operation of CC critically presupposes an up-to-date situation awareness (SA), defined as the perception of elements in the environment within a volume of time and space, the comprehension of their meaning, and the projection of their status in the near future (Endsley, 2001). SA depends on the driver's active monitoring of incoming environmental information.

Monitoring is the process of checking the task over time for quality control and the adjustment of behaviour (Stuss & Alexander, 2007). Monitoring encompasses multiple functional dimensions. At the neural level, fMRI evidence has established that the frontoparietal cortical network supports the monitoring of target events unfolding over time, constituting the neural substrate of sustained attentional control (Tarantino et al., 2017). Action monitoring is another dimension, subserved by the inferior parietal lobule (IPL) and in particular the temporoparietal junction (TPJ), which evaluates the correspondence between motor intentions and perceived sensory outcomes or feedback (Balslev et al., 2006). In the driving context, evidence from eye-tracking studies shows that as automation level increases and motor agency decreases, drivers progressively reduce

sampling of task-critical areas such as the near road surface and distant traffic (Deniel et al., 2024). Beyond this visual narrowing, the shift toward passive supervision renders drivers susceptible to broader attentional disengagement from the driving scene, which is a process closely linked to the vigilance decrement (Navarro, 2019). Compounding this, the sustained demand for purely visual-cognitive monitoring without active motor engagement gives rise to passive fatigue, an underload-induced state that further erodes alertness (Saxby et al., 2013). This failure of the monitoring caused the Out-of-the-Loop (OOTL) problem (Endsley, 1995; Gouraud et al., 2017). As a direct consequence of this cognitive and situational disengagement, the driver needs more cognitive resources to mount the rapid, goal-directed response when confronted with a sudden critical event such as emergency braking by a lead vehicle, thereby imposing a substantially greater cognitive switching cost at the moment of takeover.

Task switching has been characterized as switching back and forth between multiple tasks, operations, or mental sets (Miyake et al., 2000). Mental set-shifting tasks require multiple subprocesses, including interference control, retrieval of task sets, and task-set reconfiguration (Monsell, 2003). A behavioral signature of this process is the switch cost : In mixed-task paradigms, participants consistently respond slower in switch trials compared to repetition trials. This reaction time cost reflects the cognitive resources allocated to advance task preparation and the suppression of proactive interference from prior task sets (e.g., Koch, 2001; Meiran, 1996; Rogers & Monsell, 1995). Task switching shares a common prefrontal basis with CC and involves a process-specific cognitive mechanism. Functional neuroimaging studies highlighted the right inferior junction and posterior parietal cortex as domain general zones for switching (Esterman et al., 2009; Kim et al., 2011). Ravizza and Carter (2008) identified two dissociable forms of task switching with distinct neural substrates. Defined as updating the response rule applied to a given stimulus, rule switching is positively associated with activation in the dorsolateral

prefrontal cortex (DLPFC). Of greater relevance to the driving context is perceptual switching, which involves reorienting attention across features or spatial locations and relies specifically on the right superior parietal cortex (rSPC). High and low performing switchers exhibit comparable levels of activation in this region, suggesting that rSPC engagement reflects the engagement of the switching process itself. In the context of HAD, switch costs are amplified, encompassing both cognitive and physical dimensions. At the cognitive level, due to the OOTL phenomenon, drivers must mentally reconfigure their role from passive supervisor to active operator upon resuming manual control. Consequently, an additional decisional layer is introduced, the driver must first decide to override the system and return to manual control before executing a corrective manoeuvre, potentially introducing response delays (Navarro et al., 2011). At the physical level, meta-analytic evidence indicates that even the isolated motor transition of moving one's hands from a non-driving activity back to the steering wheel is sufficient to increase take-over time by approximately 1.33 seconds (Zhang et al., 2019). Thus, a TOR cannot be adequately characterized as a purely cognitive event. It constitutes a multi-layered transition engaging perceptual, cognitive, and motor processes in a tightly coupled loop.

Eye-hand coordination constitutes the primary operational mechanism through which fine motor control is achieved during driving. Navarro et al. (2020) demonstrated that drivers exhibit predictive gaze behavior during steering, in which ocular displacement anticipates the future vehicle trajectory, providing feedforward input for the brain's internal model to compute upcoming motor commands. Building on this, Deniel et al. (2024) established the foundational role of this visuomotor cooperation in active driving. When automated driving eliminates the demand for fine motor execution, gaze behavior undergoes a qualitative shift, regressing from a proactive, motor-coupled scanning strategy toward a passive monitoring pattern. When a TOR subsequently compels drivers to prepare for the resumption of physical control, a compensatory visual strategy appears to emerge at

the behavioral level, which may reflect an attempt to urgently integrate the environmental parameters necessary to reactivate the dormant sensorimotor circuit and re-establish the perceptual foundation for SA. This evidence suggests that active motor engagement is not merely an output of the driving task, but a structuring force that organizes the entire perceptual-cognitive loop.

Despite extensive mapping of the neural substrates underlying cognitive control, monitoring, switching, and motor control, a critical limitation remains : studying these sub-processes in isolation obscures how they are dynamically integrated in complex real-world scenarios (Benn et al., 2014). In the context of automated driving, these four functions must interact within a tightly coupled, time-critical loop, throughout the complete take-over cycle. To address these lacunae, the present study investigates the dynamic neural and behavioral impacts of four automation levels across the three critical phases of the take-over process. Functional near-infrared spectroscopy (fNIRS) offers a suitable solution in this context.

1.2 Hypotheses

We hypothesize that the level of driving automation (LoA) systematically modulates drivers' cortical hemodynamic responses across the three PoI, with the magnitude of these responses showing a linear relationship with LoA.

First, during the pre-TOR phase (nominal assisted driving, NDP), increasing LoA is expected to induce a progressive disengagement from manual control and active task monitoring. This disengagement should manifest as a downward trend in the task-evoked HbO response over cortical regions associated with motor control (overlying the supplementary motor area, premotor cortex, and primary motor cortex), as well as over fronto-parietal regions supporting monitoring and cognitive control. This pattern would reflect a shift from active manual control toward supervisory processing.

Second, during the take over preparation period (TOPP), higher LoA is expected to impose greater cognitive demands associated with task switching. Specifically, we predict that the neural switch cost, operationalized as the increase in the HbO response from the NDP to the TOPP phase, will show an upward trend with LoA, expressed as stronger recruitment of cortical regions overlying the dorsolateral prefrontal cortex and associated monitoring networks. This effect is driven by the need to rapidly reconfigure task sets and re-engage with the driving environment following prolonged disengagement under higher automation.

Third, in the post TOR manual driving period (MDP) , higher LoA is expected to render the re-engagement of motor and cognitive processes less efficient. We propose that the elevated switch cost incurred at higher LoA carries over into the post-take-over phase, such that re-establishing situational awareness and manual control demands greater and more sustained neural effort. This inefficiency is expected to manifest as an amplified task-evoked HbO response over motor control and cognitive control regions, with a linear relationship to LoA. Rather than reflecting a simple return to baseline, this amplified response would index the additional processing required to overcome automation-induced disengagement and to reconstruct an accurate driving situation.

Finally, we hypothesize that the magnitude of the hemodynamic changes between successive task phases (inter-POI transitions) will itself show a linear relationship with LoA. Higher levels of automation are expected to amplify the rise in switch-related activity at TOPP relative to NDP, and the subsequent re-engagement-related activity (cognitive control, monitoring and motor control) at MDP relative to NDP and TOPP.

2. Method

A driving simulator experiment was conducted in which different AL were manipulated in a non-critical driving situation. Four levels of automation (LoA) were

defined along from fully manual to fully automated control, and all participants were exposed to each LoA condition.

2.1. Participants

Participants were contacted through social networks or in person, most of them are at the Université Lumière Lyon 2 campus. Holding a driving licence above one year and not suffering from transport/simulation sickness were the only inclusion/exclusion criteria. After receiving information about the experiment and signing an informed consent, volunteers were requested to fill out a form to provide driving experience information (cf. Annexe 2).

A total of 71 volunteers participated in the experiment (60 females, 11 males), ranging in age from 19 to 45 years ($M = 24.35$, $SD = 4.78$), with a mean driving experience of 4.3 years; two participants reported more than 20 years of experience. Due to technical issues during data acquisition, data from 37 participants were lost and excluded from analysis. The final sample comprised 34 participants (aged 20–45 years, 29 females and 5 males; $M = 24.17$, $SD = 5.83$), with a mean driving experience of 4.97 years. Self-reported annual driving distance averaged 6844.35 km ($SD = 7295.1$). Of these, 18 participants reported driving fewer than 5,000 km/year, 13 between 5,000 and 15,000 km/year, and the remaining 3 more than 15,000 km/year.

2.2. Apparatus

The fixed-based driving simulator used in this study was developed at the Université de Sherbrooke. Simulation was displayed on a three-screens setup (48 cm × 30 cm each, covering a total of 137.52° of the visual angle on the horizontal axis and 28.65° of the visual angle on the vertical axis). Drivers' inputs were collected via a Logitech G29 steering wheel and pedal device. The driving seat was adjustable, and the simulated vehicle was set on automated gear mode.

During the study, the hemodynamic activity was recorded with a Photon Cap C20 fNIRS system (made by Cortivision). We used 16 emitters and 10 detectors and allowed a total 39 channels of measurement. Four of these were short-separation channels (source-detector distance ≈ 10 mm), which were used to regress out systemic physiological contributions and were not assigned to any region of interest. The remaining 35 long-separation channels (≈ 30 mm) were used for the cortical analyses and grouped into four functional ROIs. The spatial layout of the emitters and detectors, together with the channels grouped into regions of interest (ROIs), is shown in Figure 1 (A). and the projection of the channel positions onto the surface of the brain are shown in Figure 1 (B). The emitters use optical signals of waves of various lengths, emitted simultaneously by each source. The data were calibrated at the start of the experiment and recorded using the Cortiview software.

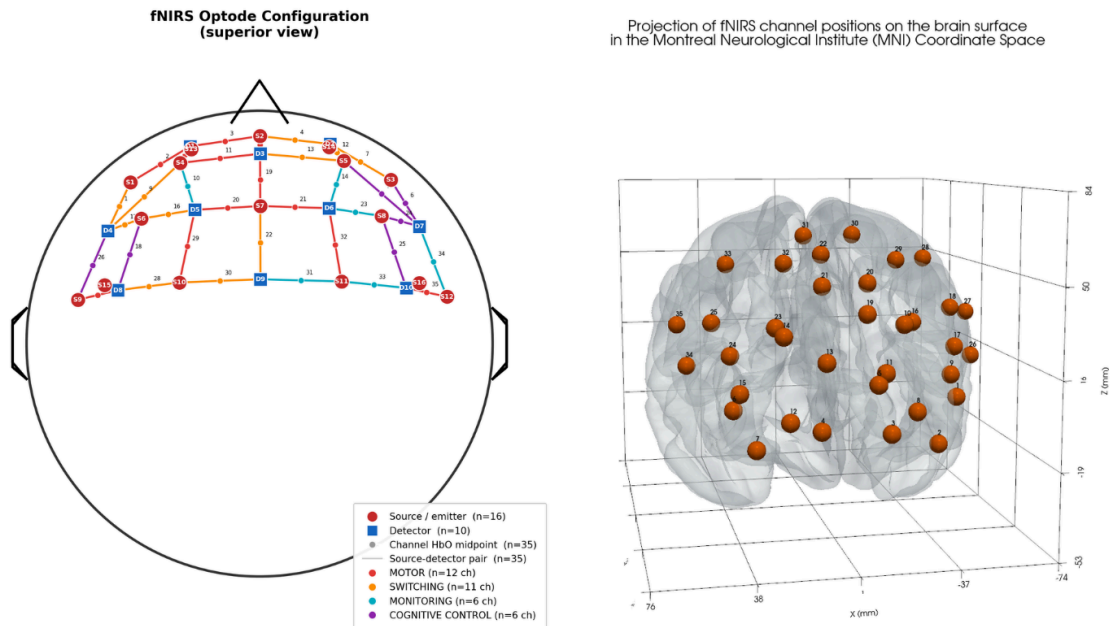


Figure 1 (A). fNIRS optode montage over the superior view. Red circles denote sources ($n=16$), blue squares denote detectors ($n=10$). Each long-separation channel is represented by its source-detector midpoint ($n = 35$), colour-coded by region of interest: motor (red), switching (orange), monitoring (cyan), and cognitive control (purple). Figure 1 (B). fNIRS channel positions projected onto the MNI fsaverage brain surface (pial) and numbered consistently with Figure 1 (A). Red spheres represent the 35 HbO channel midpoints.

2.3 Stimuli

The experimental stimuli comprised three categories: dynamic traffic scenes, physical road hazards, and take-over alerts. **Traffic Scene Stimuli.** The baseline visual environment consisted of 5 distinct environment types: rural roads, 2 different highways, urban streets, and mountain roads, providing ecological validity across varied driving contexts. **Obstacle Stimuli.** The critical event sequence, serving as the primary trigger for manual intervention, included a roadside warning sign followed by a scenario requiring the driver to perform a lane-change maneuver. Specific obstacle scenarios included a stationary light truck on the roadside, road maintenance works, improperly parked vehicles, and a highway exit requiring the driver to resume manual control to navigate the exit ramp. **TOR Stimuli.** The TOR was delivered exclusively through the visual channel. Seven seconds prior to the warning sign, a countdown timer appeared in the lower-right corner of the central dashboard display, decreasing second by second. Upon countdown completion, the display showed the label "Off", constituting the final stimulus signaling the transfer of vehicle control to the driver.

2.4 Variables :

Independent Variables :

Automation systems and associated automation levels. Four LoA were manipulated with the use of four different automation solutions. This four level LoA gradation aligns with the engagement continuum proposed by Merat et al. (2019) through the "in-/on-/out-of-the-loop" framework. Treating the driver as the primary agent, LoA decreases as the degree of automation increases, such that LoA1 represents the highest level of driver engagement and LoA4 the lowest. (1) LoA1 : Full Manual Control (Man; SAE Level 0). In LoA1, the driver had full responsibility for vehicle control, operating both the steering wheel and pedals without any automation assistance. This condition represents the highest level of automation and serves as the reference baseline. (2) LoA2 :

Adaptive Cruise Control (ACC; SAE Level 1). LoA2 introduced longitudinal automation : vehicle speed was managed by the system while the driver retained full responsibility for steering. This condition required a slightly reduced level of motor engagement relative to LoA1. (3) LoA3 : Shared Control (SC; SAE Level 2). LoA3 involved a medium level of motor engagement through a shared control paradigm. Speed was handled by the automation, while lateral control was shared: the driver was required to steer actively, but the system applied a proportional resistive force to the steering wheel whenever a trajectory deviation was detected. This haptic feedback discouraged over-reliance on the automation, as steering input remained necessary to maintain lane position. (4) LoA4 : Highly Automated Driving (HAD; SAE Level 3). LoA4 represented the lowest level of motor engagement. HAD removed all steering wheel motion feedback, and drivers were explicitly instructed to relinquish vehicle control entirely to the automation, with no requirement to interact with or maintain contact with the steering wheel.

Periods of interest. The effects of these LoA were examined across three predefined periods of interest (PoI): (1) the Nominal Driving Period (NDP), during which participants drove under active automation in a steady-state context; (2) the Take-Over Preparation Period (TOPP), initiated upon issuance of a TOR, capturing the interval during which drivers prepared to resume manual control; and (3) the Manual Driving Period (MDP), encompassing the initial moments following the resumption of manual vehicle control. The driving context, key events, and PoI are illustrated in Figure 2.

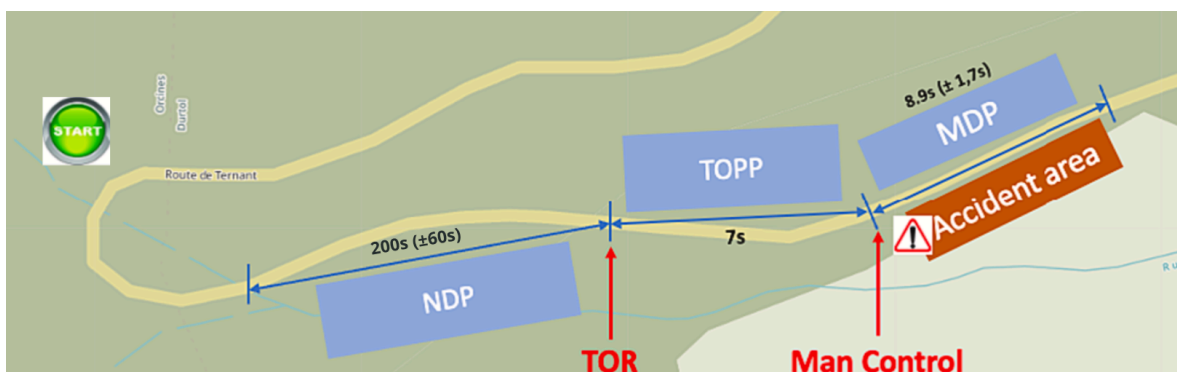


Fig. 2. Illustration of the periods of interest for the considered driving situation. NDP: Nominal Driving Period; TOR: Take-over Request; TOPP: Take-Over Preparation Period; Man Control: manual control resuming point; MDP: Manual Driving Period.

Dependent Variables. The dependent variable is the task-evoked cortical hemodynamic response, operationalized as the activation coefficient (θ) for oxygenated hemoglobin (HbO), estimated via a general linear model (GLM). The coefficient θ thus indexes the amplitude of the task-related component of the ΔHbO signal rather than the raw concentration change itself. Larger values of θ reflect a stronger task-evoked hemodynamic response and, by inference, greater neural and metabolic demand in the underlying cortical tissue.

2.5 Procedure

Following informed consent and fNIRS equipment fitting and calibration, participants completed a training session of approximately 30 minutes. The session began with a free-driving familiarization phase to ensure full proficiency with the simulator's physical controls. The experimenter then systematically introduced and demonstrated each of the four LoA in turn. Participants were informed that they could not disengage the automation system independently, and that all control transfers were contingent upon a system-issued take-over request (TOR).

During the experimental phase, participants completed the driving scenarios in a counterbalanced order (4 LoA $\times$ 5 situations). Table 1. describes the four different orders of presentation defined. Prior to each scenario, participants received only a brief verbal indication of the current automation type. No visual icon indicating the active automation level was displayed on the HMI, in order to minimize additional visual distraction.

Table 1. Partial counterbalanced order of the levels of automation.

	Run 1	Run 2	Run 3	Run 4
Order 1	HAD	SC	ACC	Man
Order 2	SC	ACC	Man	HAD

Order 3	ACC	Man	HAD	SC
Order 4	Man	HAD	SC	ACC

2.6 Data Analysis

fNIRS data were preprocessed in Python using MNE-NIRS following a standard pipeline. Motion artifacts in the raw optical density signals were first corrected using a wavelet-based approach. The corrected signals were band-pass filtered between 0.01 and 0.1 Hz to remove low-frequency drift and high-frequency physiological noise, then converted into concentration changes of oxygenated (ΔHbO) and deoxygenated hemoglobin (ΔHbR) using the modified Beer–Lambert law. Given the higher signal-to-noise ratio and greater sensitivity of ΔHbO to task-related hemodynamic changes, all subsequent analyses were conducted on the 35 ΔHbO channels.

To quantify task-evoked cortical activation, a channel-level general linear model (GLM) was implemented to estimate β coefficients serving as indices of activation strength. Driving periods across the four levels of automation (LoA: MAN, ACC, SC, and HAD) were modeled as task-related regressors. To capture distinct phases of the takeover process, three periods of interest (PoI) were defined relative to the takeover request (TOR). The normal driving period (NDP) was defined as the 15-s window from 30 to 15 s before the TOR. The takeover preparation period (TOPP) was defined as the 7 s window immediately following the onset of the TOR signal. The manual driving period (MDP) was defined as the 15 s window beginning at the end of the TOPP. Each regressor was convolved with a canonical hemodynamic response function to generate the model predictors.

At the region-of-interest (ROI) level, hypothesis testing was conducted in R using linear mixed-effects models (lme4) with estimated marginal means and contrasts (emmeans). Channels were grouped into functional ROIs corresponding to motor control,

task-switching, monitoring, and cognitive control. ΔHbO activation was modeled as a function of ROI, PoI, and LoA, with their full three-way interaction and a by-subject random intercept. A model additionally including by-subject random slopes for PoI was initially specified but yielded a singular fit; the random-intercept structure was therefore retained. The levels of LoA were ordered from lowest to highest automation (MAN < ACC < SC < HAD), and the test of each prediction was the linear polynomial trend across LoA. Within each ROI $\times$ PoI combination, the estimated marginal means of the four LoA levels were submitted to orthogonal polynomial contrasts (linear, quadratic, cubic): a significant linear term indexed a monotonic increase (positive estimate) or decrease (negative estimate) in activation with automation, whereas significant higher-order terms were used to flag departures from strict monotonicity. Tukey-corrected pairwise comparisons between LoA levels were used in addition to localize which levels differed. To characterize the transitions between PoI, the estimated marginal means were contrasted across PoI within each LoA $\times$ ROI cell (reverse-pairwise contrasts: TOPP – NDP, MDP – NDP, MDP – TOPP), giving the magnitude of each phase difference. The test of whether these phase differences themselves scaled with automation was an interaction contrast crossing the reverse-pairwise PoI differences with the linear LoA trend, evaluated separately within each ROI.

3. Results

Overall model. A linear mixed-effects model was fitted to the task-evoked HbO response (θ) with ROI, LoA, and PoI as fixed effects and a by-participant random intercept. Results revealed a main effect of PoI, $F(2, 1231) = 17.18$, $p < .001$, and a significant LoA $\times$ PoI interaction, $F(6, 1231) = 4.03$, $p < .001$. The ROI $\times$ PoI interaction reached only marginal significance, $F(6, 1231) = 1.90$, $p = .077$. Neither the main effects of ROI, $F(2, 1231) = 0.97$, $p = .41$, or LoA, $F(2, 1231) = 0.25$, $p = .86$, nor the three-way ROI $\times$ LoA $\times$ PoI interaction, $F(18, 1231) = 0.87$, $p = .62$, were significant. **PoI main effect.**

Averaging over ROI and LoA, the MDP was significantly lower than both the NDP ($\Delta = -0.34$, 95% CI $[-0.50, -0.19]$, $t = 5.17$, $p < .001$) and the TOPP ($\Delta = -0.32$, 95% CI $[-0.47, -0.16]$, $t = 4.76$, $p < .001$), whereas the NDP and TOPP did not differ ($p = .91$; cf. Fig 3 (A)).

The PoI $\times$ LoA interaction effect. The significant LoA $\times$ PoI interaction indicates that the effect of PoI on θ depends on the LoA (cf. Fig 3 (B)). Notably, LoA produced no reliable main effect, indicating that its influence was PoI-specific. **Simple effects of PoI within each LoA.** Averaging over ROI, we observed a significant decline in θ during the MDP phase compared to both the NDP and TOPP phases (MDP-NDP: ACC: $\Delta = -0.63$, $p < .0001$; HAD: $\Delta = -0.67$, $p < .001$; MDP-TOPP: ACC: $\Delta = -0.39$, $p < .001$; HAD: $\Delta = -0.53$, $p < .001$).

Simple effects of LoA within each PoI. Averaging over ROI, we observed that, during the NDP phase, θ was significantly higher in the ACC condition compared to both the MAN and SC conditions (ACC - MAN: $\Delta = 0.37$, $p < .05$; ACC-SC: $\Delta = 0.37$, $p < .05$). During the MDP phase, θ was significantly higher in the MAN condition compared to the HAD condition ($\Delta = 0.40$, $p < .05$).

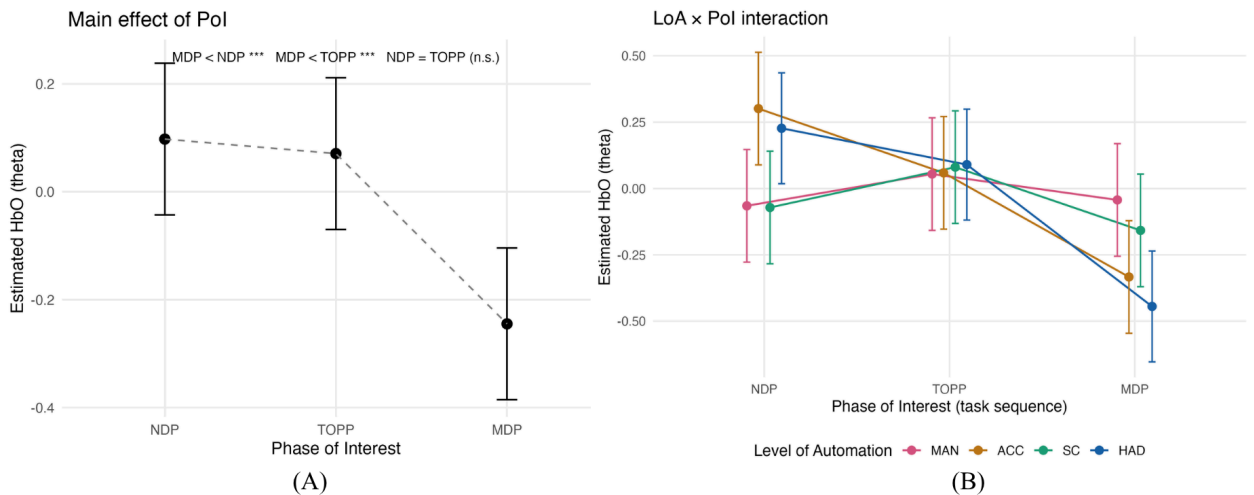


Figure 3. Effects of PoI and LoA on the task-evoked HbO response (θ). All panels display estimated marginal means from the linear mixed-effects model; error bars denote 95% confidence intervals. (A) Main effect of PoI (averaged over ROI and LoA). θ was significantly lower in the MDP than in both the NDP and the TOPP (both $p < .001$), whereas the NDP and TOPP did not differ ($p = .91$). (B) LoA $\times$ PoI interaction (averaged over ROI). θ is plotted across the three PoI for each LoA. The LoA $\times$ PoI interaction is significant ($F = 4.03$, $p < .001$).

Exploratory ROI Analysis.

Based on our a priori hypotheses regarding specific functional brain regions, we conducted further analyses to examine the simple effects of PoI and LoA within each

region of interest (ROI). **Simple effects of PoI within each LoA × ROI.** PoI contrasts were examined within each LoA × ROI combination (cf. Fig 4). From NDP to MDP, for both the HAD and the ACC LoA condition, motor control and cognitive control showed a decrease (HAD : motor control: $\Delta = -0.77$, $p < .01$; cognitive control: $\Delta = -0.96$, $p < .001$; ACC : motor control : $\Delta = -1.07$, $p < .001$; cognitive control : $\Delta = -1.18$, $p < .01$). Also, under the HAD condition, the switching region also showed a decrease ($\Delta = -0.80$, $p < .01$). From TOPP to MDP, for both HAD and ACC conditions, only motor regions showed a significant decrease (HAD : $\Delta = -0.75$, $p = .001$; ACC : $\Delta = -0.77$, $p = .011$). **Simple effects of LoA within each PoI × ROI.** LoA contrasts within each PoI × ROI combination revealed an effect that only in the MDP and only the cognitive control region differed across automation levels (MAN – ACC : $\Delta = 0.77$, $p = .02$; MAN – HAD : $\Delta = 1.00$, $p < .01$; cf. Fig 5).

Linear modulation of inter PoI difference by LoA within each ROI. Further, to test whether the signed inter-PoI differences varied with LoA, an interaction contrast crossing each PoI difference with the linear LoA trend was computed. The MDP – NDP difference showed a negative linear relationship with LoA in the cognitive control region ($t(1282) = -2.33$, $p = .02$) and the switching region ($t(1282) = -2.29$, $p = .022$), because this difference was itself negative, the relationship indicates that the magnitude of the MDP reduction relative to NDP grew larger as automation increased. The MDP-TOPP difference also showed a negative linear relationship with LoA in the cognitive control region ($t(1282) = -2.12$, $p = .03$). No corresponding linear relationship with LoA was found in the monitoring or motor regions (all $p > .21$). The TOPP – NDP difference showed no significant linear relationship with LoA in any ROI. **Linear trends in within-PoI θ across LoA within each ROI.** Linear trend analyses converged on the same ROI. No reliable linear relationship with LoA showed in the NDP or TOPP for any ROI (all $p > .17$) : during the NDP the motor, monitoring, and cognitive control regions were all

flat (motor: $t(1282) = 1.09, p = .28$; monitoring: $t(1282) = -0.04, p = .97$; cognitive control: $t(1282) = 0.03, p = .98$), and during the TOPP the switching region was flat ($t(1282) = -0.93, p = .35$). In these phases, the slopes were not distinguishable from zero. In the MDP, by contrast, the cognitive control region showed a negative linear relationship with LoA ($t(1282) = -3.26, p = .001$; cf. Fig. 5), with θ declining as automation increased. The switching region showed a comparable downward trend that approached significance ($t(1282) = -1.90, p = .057$), whereas the monitoring and motor regions showed none (both $p > .36$). A significant cubic component in the cognitive control region ($p = .03$) indicated that this decline was not perfectly linear but carried some curvature.

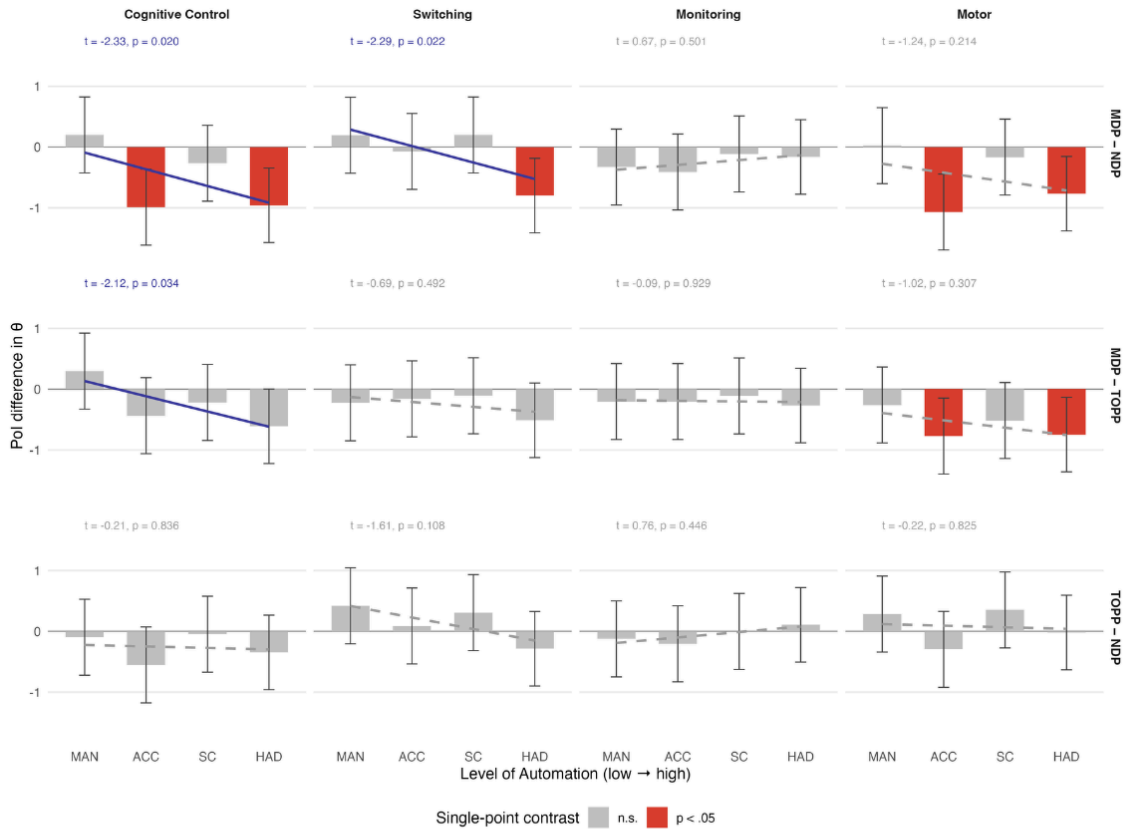


Figure 4. Pol differences in the task-evoked HbO response (θ) at each level of automation, by ROI.

Bars show the estimated Pol difference in θ from the linear mixed-effects model, computed within each region of interest at each level of automation (x-axis, ordered from low to high). Negative values indicate that θ was lower in the MDP than in the reference phase. Bar colour indicates whether that individual contrast was significant (red, Tukey-adjusted $p < .05$; grey, n.s.); error bars denote 95% confidence intervals (± 1.96 SE). **The superimposed straight line** shows the linear trend of the Pol difference across automation level. Solid blue line means a significant linear trend and a dashed grey line represent a non-significant linear LoA trend; the t and p value of this trend are annotated in each panel.

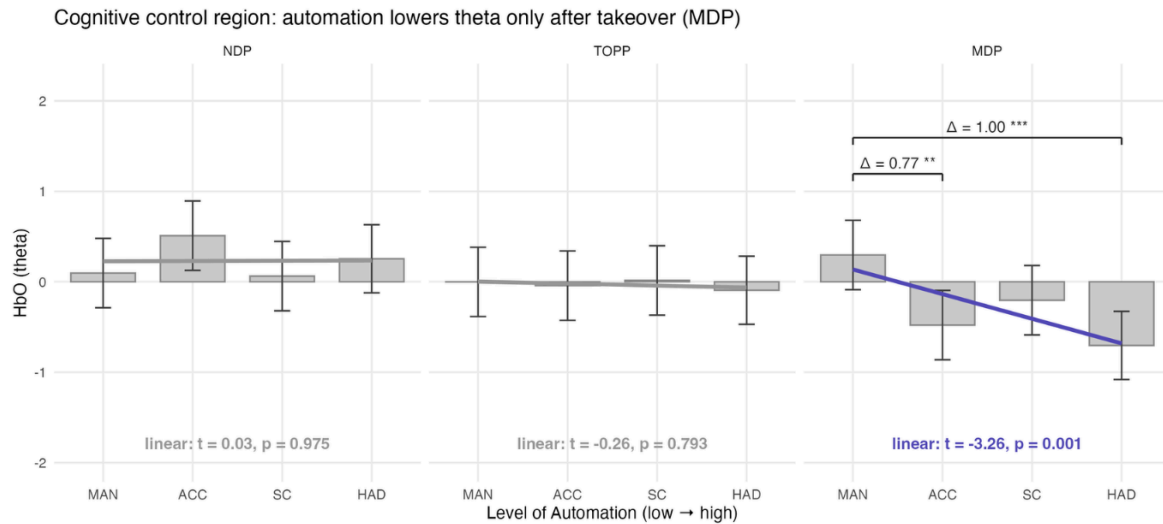


Fig. 5. θ in the cognitive control region as a function of level of automation, shown separately for the three PoI. Bars are model-estimated marginal means ($\pm 95\%$ CI) from the full model. Brackets denote significant pairwise LoA contrasts (** $p < .01$, *** $p < .001$). Only the cognitive control ROI is shown because, across all PoI $\times$ ROI combinations, it was the only region in which automation level modulated θ , and only in the MDP. The overlaid line is the linear LoA trend, coloured purple where significant ($p < .05$) and grey otherwise, with t and p annotated.

4. Discussion

4.1 Interpretation of the results

The present findings demonstrate that the phase of interest (PoI) was the dominant modulator of task-evoked HbO responses, with the MDP phase eliciting significantly lower θ than both the NDP and TOPP phases. The absence of a difference between NDP and TOPP suggests that this attenuation was specific to the MDP phase, pointing to a reduction in cortical engagement within the sampled regions following manual takeover. Automation did not exert a uniform influence on cortical activity but instead shaped responses in a PoI-dependent manner. During the NDP phase, θ was elevated under the ACC condition relative to both the MAN and SC conditions, indicating that both longitudinal automation (relative to manual control) and lateral control (relative to SC) enhanced HbO in the sampled regions. During the MDP phase, by contrast, θ in the MAN condition exceeded that in the HAD condition, suggesting that cortical responses following manual takeover continued to be modulated by the preceding level of automation. Finally, while the model revealed no robust ROI main effect, this does not preclude region specific patterns that a more targeted analysis might uncover.

In the cognitive control region, activity following takeover (MDP) was significantly lower than during both continuous driving (NDP) and the takeover-preparation phase (TOPP), under both the HAD and ACC automation levels. Crucially, the magnitude of this post-takeover reduction grew systematically with the level of automation: within the MDP, cognitive control activity was lower under both ACC and HAD than under manual driving and declined as automation increased, following a predominantly linear trend with some additional curvature. Together, these patterns indicate that the higher the level of automation, the greater the post-takeover disengagement of the cognitive control region. The motor control region showed a different profile. MDP activity was again significantly lower than during both NDP and TOPP, under both HAD and ACC conditions, but this region was insensitive to the level of automation, exhibiting no linear trend. Thus, under the present paradigm, motor control disengaged after takeover regardless of how much automation had preceded it. The switching region displayed an intermediate, marginal pattern. MDP activity fell below NDP only under the HAD condition, and the within-MDP trend across automation levels only approached significance. We interpret this as a pattern resembling that of the cognitive control region, but in attenuated form. Finally, the monitoring region showed no reliable effect in any phase or across any automation comparison. Taken together, we found no global or sustained change in the measured regions as a function of automation; rather, the effects were confined to specific regions at specific phases of the driving task. Automation's modulation of post-takeover neural activity was therefore region-specific and cognitive-control-dominated.

4.2 Theoretical implications

Contrary to the effortful re-engagement we had hypothesized, we found that the deeper the preceding automation, the less complete the re-engagement of the cognitive-control system following takeover. We interpret this as the cost of disengagement

rather than of recovery. The OOTL framework (Endsley, 1995) holds that automation shifts the operator from an active information processor to a passive monitor, and that the deeper the automation, the more complete this passive shift. As a result, the cognitive-control processes that sustain task engagement are weakened and situation awareness is degraded. Although this state has largely been characterized as a phenomenon arising during and immediately after automation, the present study extends it to the period following takeover. The under-engagement of the cognitive-control system persists once manual control has been resumed, beyond the 7 s takeover-preparation phase, and decreases linearly with the preceding level of automation. We therefore propose that post-takeover cognitive-control re-engagement is not a discrete switch but a graded, incompletely recovered process modulated by the dose of prior automation. The switching region, commonly regarded as a component of the cognitive control system, showed the same trend, reinforcing the view that this modulation was specific to higher-order control rather than reflecting a global change across regions.

The motor control region presents a more counterintuitive pattern: it too showed reduced MDP activity relative to both continuous driving (NDP) and the takeover preparation phase (TOPP). Even under the Manual condition, in which drivers operated the vehicle throughout, the task-evoked HbO response in the motor control region was negative. Because neural activation typically corresponds to an increase in HbO, this direction was unexpected, and its neurophysiological significance remains unclear. It may reflect a relative deactivation of this region during skilled manual operation, or it may instead arise from extracerebral blood-flow changes during manual driving, motion-related artefacts, or the influence of global physiological noise on the HbO channels.

5.Conclusion

The present study employed fNIRS to characterize how four levels of automation jointly modulate the task-evoked cortical hemodynamic response of the motor control,

switching, monitoring, and cognitive control networks across the three temporal periods of the take-over cycle. We found that automation's influence on post-takeover neural activity is region-specific and cognitive-control-dominated rather than global, with cognitive control activity following manual takeover declining linearly as the preceding level of automation increased, revealing the cost of automation-induced disengagement. Beyond the driving context, this graded under-engagement speaks to a broader question: how prolonged underload reshapes the state of the cognitive control system. It suggests that re-engaging top-down control is not a discrete switch flipped on demand, but a gradual process whose recovery is constrained by the depth of the preceding disengagement. This is a principle likely to extend to any setting where humans supervise automation before being called to act. As automation becomes pervasive across high-stakes domains, the question of how depth of automation shapes the human's residual capacity to act and how long that cost lingers, becomes a central concern for cognitive science and system design alike.

5.1 Limits

Several limitations qualify the interpretation of the present findings.

First, the study lacks behavioral data. No physical performance metric (e.g., take-over time, steering or braking reaction) was measured to objectively present whether a take-over was successful, and no behavioral marker of low activation was recorded. Consequently, the reduced cognitive control activation we observe cannot be fully anchored to the situation awareness construct central to the OOTL framework, nor can the participants' underlying cognitive state be interpreted at the behavioral level. The neural disengagement we report therefore remains, for now, without behavioral validation. Second, and most importantly, the statistical power of the study was constrained by an insufficient sample size. Although 71 volunteers were initially recruited, technical problems during data acquisition led to the loss of 37 datasets, leaving a final sample of

only 34 participants. This attrition substantially reduces the sensitivity of the linear mixed-effects models to detect smaller or more distributed effects, and likely contributes to the marginal patterns observed in the switching region and the absence of effects in the monitoring region. The sample was also strongly imbalanced in terms of sex, limiting the generalizability of the results. Finally, the ecological validity of the paradigm is limited. A fixed-based driving simulator cannot reproduce the vestibular and proprioceptive cues, the perceived risk, or the motivational stakes of real world driving, all of which may modulate cognitive control and motor engagement during a genuine take-over.

5.2 Perspectives

These limitations point directly toward several avenues for future research.

First, a larger, sex-balanced sample would clarify whether our findings are robust, and would adjudicate whether the marginal switching-region trend ($p = .057$) and the null monitoring effect reflect genuine functional dissociations or merely insufficient power. Second, pairing the fNIRS measures with take-over time, lateral/longitudinal control quality, and gaze-sampling would allow the observed cognitive-control disengagement to be tested as a predictor of objective take-over performance. Establishing this neuro-behavioral link is what would distinguish a genuine cost of disengagement from a benign re-allocation of resources.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT to refine the writing of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the article.



INSTITUT DE
PSYCHOLOGIE



MASTER 1^{RE} ET 2^{ME} ANNÉE MENTION SCIENCES COGNITIVES

Charte d'usage responsable de l'intelligence artificielle pour la réalisation du mémoire de recherche

Préambule

L'usage des outils d'intelligence artificielle (IA) est parfaitement légitime lorsqu'il s'inscrit dans une démarche réflexive et critique pour *apprendre mieux*. Tout usage de l'IA doit cependant rester continuellement sous votre supervision et être renseigné avec transparence.

La présente charte a pour objectif de définir un cadre d'usage responsable des outils d'IA dans le cadre de l'écriture de votre document de synthèse et de votre mémoire de recherche, ainsi que dans la préparation de vos présentations et soutenances orales, durant vos deux années de Master mention Sciences Cognitives.

Article 1 – Principe général

L'usage des outils d'IA est utile à condition qu'il soit raisonné et qu'il reste toujours sous votre contrôle vigilant. Vous êtes entièrement responsable du contenu que vous décidez d'utiliser et d'intégrer dans vos documents. Cela garantit que votre usage des outils d'IA, dans le cadre de la réalisation de votre mémoire de recherche, reste compatible avec les principes d'intégrité scientifique et d'éthique de la recherche.

Article 2 – Usages légitimes

L'utilisation d'outils d'IA est considérée comme légitime lorsqu'elle vise à favoriser votre réflexion et la clarté de votre rédaction, sans se substituer à votre expertise scientifique personnelle. Par exemple, il est possible d'utiliser une IA pour clarifier sa pensée, défendre un argument, reformuler un texte jugé insatisfaisant, compiler des références bibliographiques sur une question particulière, être conseillé sur la meilleure façon de présenter ou de choisir une analyse statistique de données, à condition de vérifier et de valider chaque production de l'IA avant son utilisation dans votre document.

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L'utilisation d'outils d'IA est considérée comme inappropriée lorsqu'elle transgresse les principes d'intégrité scientifique qui impliquent que toute production intellectuelle doit être le fruit d'un travail personnel fondé sur la vérification, la traçabilité des sources et l'honnêteté dans la présentation des résultats. Ainsi, vous ne pouvez pas utiliser un outil d'IA pour produire du contenu sans vérification et contrôle critique, pour rédiger à votre place votre mémoire ou pour manipuler vos données dans le but d'en orienter l'interprétation.

Article 4 – Transparence

Tout recours à un outil d'IA dans l'écriture de votre mémoire de recherche doit être indiqué en fin de mémoire, dans la section prévue à cet effet juste avant la section "Références", dans un paragraphe intitulé *Usage d'outils d'intelligence artificielle*. Les formulations à utiliser et la nature des informations à rapporter vous seront présentées en TD *Pratique de la Recherche*. Vous devrez vous y conformer scrupuleusement. Pour le document de synthèse et les présentations orales, la signature de cette charte tient lieu de déclaration.

Article 5 – Engagement de l'étudiant / de l'étudiante (cette charte sera signée en M1 et en M2)

Je déclare avoir pris connaissance de la présente charte et m'engage à en respecter les règles pour toutes les productions liées à la réalisation de mon mémoire de recherche.

Nom : CHEN Prénom : Shenglan

Fait à Bron, le 27/05/2026

Signature de l'étudiant(e) : Shenglan Chen

Nom du responsable pédagogique du mémoire de recherche, année 2025-2026 : **Hanna CHAINAY**

References

- Balslev, D., Nielsen, F. Å., Lund, T. E., Law, I., & Paulson, O. B. (2006). Similar brain networks for detecting visuo-motor and visuo-proprioceptive synchrony. *Neuroimage*, 31(1), 308-312.
- Benn, Y., Webb, T. L., Chang, B. P., Sun, Y. H., Wilkinson, I. D., & Farrow, T. F. (2014). The neural basis of monitoring goal progress. *Frontiers in Human Neuroscience*, 8, 688.
- Cohen, J. R., Gallen, C. L., Jacobs, E. G., Lee, T. G., & D'Esposito, M. (2014). Quantifying the reconfiguration of intrinsic networks during working memory. *PloS one*, 9(9), e106636.
- Deniel, J., Schnebelen, D., Reynaud, E., Ouimet, M. C., & Navarro, J. (2024). Driver's gaze behaviour before, during and after take-over manoeuvres: Influence of agentivity associated with different automation solutions. *Transportation Research Part F: Traffic Psychology and Behaviour*, 100, 181-196. <https://doi.org/10.1016/j.trf.2023.11.019>
- Diamond, A. (2013). Executive functions. *Annual Review of Psychology*, 64, 135-168. <https://doi.org/10.1146/annurev-psych-113011-143750>
- Dosenbach, N. U., Fair, D. A., Miezin, F. M., Cohen, A. L., Wenger, K. K., Dosenbach, R. A., ... & Petersen, S. E. (2007). Distinct brain networks for adaptive and stable task control in humans. *Proceedings of the National Academy of Sciences*, 104(26), 11073-11078.
- Endsley, M. R. (2001). Designing for situation awareness in complex systems. In *Proceedings of the Second International Workshop on symbiosis of humans, artifacts and environment* (pp. 1-14).
- Esterman, M., Chiu, Y.-C., Tamber-Rosenau, B. J., & Yantis, S. (2009). Decoding cognitive control in human parietal cortex. *Proceedings of the National Academy of Sciences of the United States of America*, 106(42), 17974-17979. <https://doi.org/10.1073/pnas.0903593106>
- Friedman, N. P., & Miyake, A. (2017). Unity and diversity of executive functions: Individual differences as a window on cognitive structure. *Cortex; a Journal Devoted to the Study of the Nervous System and Behavior*, 86, 186-204. <https://doi.org/10.1016/j.cortex.2016.04.023>
- Friedman, N. P., & Robbins, T. W. (2022). The role of prefrontal cortex in cognitive control and executive function. *Neuropsychopharmacology*, 47(1), 72-89. <https://doi.org/10.1038/s41386-021-01132-0>
- Gouraud, J., Delorme, A., & Berberian, B. (2017). Autopilot, mind wandering, and the out of the loop performance problem. *Frontiers in Neuroscience*, 11, 541.
- Hoc, J.-M., Young, M. S., & Blosseville, J.-M. (2009). Cooperation between drivers and automation: Implications for safety. *Theoretical Issues in Ergonomics Science*, 10(2), 135-160. <https://doi.org/10.1080/14639220802368856>
- Jurado, M. B., & Rosselli, M. (2007). The elusive nature of executive functions: A review of our current understanding. *Neuropsychology Review*, 17(3), 213-233. <https://doi.org/10.1007/s11065-007-9040-z>
- Kim, C., Johnson, N. F., Cilles, S. E., & Gold, B. T. (2011). Common and distinct mechanisms of cognitive flexibility in prefrontal cortex. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 31(13), 4771-4779. <https://doi.org/10.1523/JNEUROSCI.5923-10.2011>
- Koch, I. (2001). Automatic and intentional activation of task sets. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 27(6), 1474-1486. <https://doi.org/10.1037/0278-7393.27.6.1474>

- Meiran, N. (1996). Reconfiguration of processing mode prior to task performance. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 22(6), 1423–1442. <https://doi.org/10.1037/0278-7393.22.6.1423>
- Merat, N., Seppelt, B., Louw, T., Engström, J., Lee, J. D., Johansson, E., ... & Keinath, A. (2019). The “Out-of-the-Loop” concept in automated driving: proposed definition, measures and implications. *Cognition, Technology & Work*, 21(1), 87-98.
- Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience*, 24, 167–202. <https://doi.org/10.1146/annurev.neuro.24.1.167>
- Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The Unity and Diversity of Executive Functions and Their Contributions to Complex “Frontal Lobe” Tasks: A Latent Variable Analysis. *Cognitive Psychology*, 41(1), 49–100. <https://doi.org/10.1006/cogp.1999.0734>
- Monsell, S. (2003). Task switching. *Trends in Cognitive Sciences*, 7(3), 134–140. [https://doi.org/10.1016/s1364-6613\(03\)00028-7](https://doi.org/10.1016/s1364-6613(03)00028-7)
- Navarro, J. (2019). A state of science on highly automated driving. *Theoretical Issues in Ergonomics Science*, 20(3), 366–396. <https://doi.org/10.1080/1463922X.2018.1439544>
- Navarro, J., Mars, F., & Young, M. S. (2011). Lateral control assistance in car driving: Classification, review and future prospects. *IET Intelligent Transport Systems*, 5(3), 207–220. <https://doi.org/10.1049/iet-its.2010.0087>
- Ravizza, S. M., & Carter, C. S. (2008). Shifting set about task switching: Behavioral and neural evidence for distinct forms of cognitive flexibility. *Neuropsychologia*, 46(12), 2924-2935.
- Rogers, R. D., & Monsell, S. (1995). Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology: General*, 124(2), 207–231. <https://doi.org/10.1037/0096-3445.124.2.207>
- Sadaghiani, S., & D'Esposito, M. (2015). Functional characterization of the cingulo-opercular network in the maintenance of tonic alertness. *Cerebral cortex*, 25(9), 2763-2773.
- SAE International Recommended Practice, Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles, SAE Standard J3016_202104, Revised April 2021, Issued January 2014, https://doi.org/10.4271/J3016_202104.
- Saxby, D. J., Matthews, G., Warm, J. S., Hitchcock, E. M., & Neubauer, C. (2013). Active and passive fatigue in simulated driving: Discriminating styles of workload regulation and their safety impacts. *Journal of Experimental Psychology: Applied*, 19(4), 287–300. <https://doi.org/10.1037/a0034386>
- Stuss, D. T., & Alexander, M. P. (2007). Is there a dysexecutive syndrome? *Philosophical Transactions of the Royal Society B: Biological Sciences*, 362(1481), 901–915. <https://doi.org/10.1098/rstb.2007.2096>
- Tarantino, V., Mazzonetto, I., Formica, S., Causin, F., & Vallesi, A. (2017). The neural bases of event monitoring across domains: A simultaneous ERP-fMRI study. *Frontiers in Human Neuroscience*, 11, 376.
- Zhang, B., De Winter, J., Varotto, S., Happee, R., & Martens, M. (2019). Determinants of take-over time from automated driving: A meta-analysis of 129 studies. *Transportation Research Part F: Traffic Psychology and Behaviour*, 64, 285–307. <https://doi.org/10.1016/j.trf.2019.04.020>

Annexe 1

Le formulaire de consentement éclairé



FORMULAIRE DE CONSENTEMENT DE PARTICIPATION À UNE EXPÉRIENCE – 2025-2026

Étude : Activation cérébrale du conducteur lors de la reprise de contrôle selon le niveau d'automatisation

Investigateur EMC : Shantalou CHEN

Je soussigné :

NOM : **PRÉNOM :**

Déclare :

- que l'investigateur susnommé m'a proposé de participer à l'étude susnommée
- qu'il m'a expliqué en détail le protocole
- qu'il m'a notamment fait connaître :
 - l'objectif, la méthode et la durée de l'étude
 - les contraintes et les risques potentiels encourus
 - mon droit de refuser de participer et, en cas de désaccord, de retirer mon consentement à tout moment
 - mon droit d'interrompre ma participation à tout moment
 - qu'à terme du protocole, si je le souhaite, je serai informé par l'investigateur de ses résultats globaux
- que j'ai répondu en toute bonne foi aux questions posées
- que j'ai pu poser toutes les questions que je voulais.

Les informations relatives à l'étude susnommée, recueillies par l'investigateur, sont traitées confidentiellement, sous la responsabilité de la **Présidente de l'Université, Mme NATHALIE DOMPNIER**. J'accepte que les données enregistrées à l'occasion de cette recherche puissent faire l'objet d'un traitement informatisé anonyme, avec pour finalité une meilleure compréhension des mécanismes cognitifs impliqués dans les tâches réalisées. Les personnes physiques/morales destinataires de ces données sont le **LABORATOIRE D'ÉTUDE DES MÉCANISMES COGNITIFS (UR EMC)** et leurs représentants. J'ai bien noté que, conformément aux dispositions de la loi CNIL et au Règlement Européen sur la Protection des Données Personnelles de 25 mai 2018, je dispose à tout moment d'un droit d'accès, de portabilité, de rectification, d'effacement, de limitation et d'opposition au traitement des données me concernant. Ces droits s'exercent auprès de l'investigateur du projet.

Après avoir discuté librement et obtenu réponse à toutes mes questions, j'accepte librement et volontairement de participer à cette étude dans les conditions précisées dans le présent formulaire de consentement.

Fait à le :/...../..... en deux exemplaires originaux.

Signature du Participant / de la Participante précédée de la mention "Lu, compris et approuvé"

Annexe 2

Le questionnaire d'expérience de conduite

Questionnaire d'expérience d'automobile

Bonjour,

Je m'appelle Shenglan CHEN, étudiante en M2 Sciences Cognitives parcours Interaction Homme-Machine à l'Université Lumière Lyon 2.

Dans le cadre de mon stage de M2 au laboratoire EMC, sous la supervision de Monsieur Jordan Navarro, je réalise une étude sur l'activation cérébrale lors d'une demande de reprise en main dans un contexte de conduite automobile.

Je recrute actuellement des participants. Si cette étude vous intéresse, merci de remplir le questionnaire ci-dessous pour vous inscrire.

Les informations recueillies dans ce questionnaire sont strictement confidentielles et anonymes. Elles serviront uniquement à vérifier votre éligibilité pour l'étude.

Merci pour votre attention !

* Indique une question obligatoire

Votre code de participation : *

- les 2 premières lettres de votre NOM ;
- les 2 premières lettres de votre PRÉNOM ;
- votre jour de naissance (1 ou 2 chiffres).

Exemple :
Nom = Martin, Prénom = Julie, né e le 7 → MAJU7

Votre réponse

Votre genre *

☐ Homme

☐ Femme

☐ Autre

☐ Je ne souhaite pas répondre

Votre âge *

Votre réponse

Avez-vous le permis de conduire depuis au moins un an ? *

☐ OUI

☐ NON

Suivant

Page 1 sur 3

Effacer le formulaire

N'envoyez jamais de mots de passe via Google Forms.

Section sans titre

A quelle date avez-vous obtenu votre permis de conduire ? *

Veillez entrer la date au format AAAA-MM (ex. 2024-07)

Votre réponse

Combien de kilomètres parcourez-vous en voiture par an en moyenne ? Veuillez indiquer une estimation.

Votre réponse

Avez-vous déjà utilisé des systèmes d'assistance à la conduite ? *

☐ OUI

☐ Non

Si oui, merci de les décrire brièvement en quelques mots.

Votre réponse

Avez-vous une vision normale ou corrigée à la normale ? *

☐ OUI

☐ NON

Avez-vous des antécédents de troubles neurologiques ou de troubles psychiatriques diagnostiqués ? *

☐ OUI

☐ NON

Prenez-vous actuellement un traitement médicamenteux pouvant affecter le système nerveux ? *

☐ OUI

☐ NON

Vous êtes *

☐ Gaucher

☐ Droitier

☐ Ambidextre

Votre adresse mail *

Votre réponse

Merci pour votre participation !

Si vous n'avez pas encore de rendez-vous, une fois votre réponse envoyée, un lien de réservation s'affichera dans **le message de confirmation** pour choisir votre créneau.